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Modeling Costs of Turns in Route Planning

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Abstract

In this paper a model that handles costs of turns in route planning is defined, investigated and discussed. Costs are traditionally attached to edges in a graph. For some important route planning problems other costs can be identified, namely costs that appear when leaving one edge and entering the next. Examples are turn restrictions, the turning angle, or the simple necessity to turn. Such costs cannot be stored as attributes of nodes or edges in the graph, and they cannot be handled correctly by shortest path algorithms without modifications.

Turn costs can be represented by a pseudo-dual graph in a way that shortest path algorithms run without modifications. Although the idea is not new, it has not found much interest in the literature. The pseudo-dual graph is defined here in a new way, it is systematically investigated, and some practical applications are shown. Concentrating strictly on topology, it turns out that the pseudo-dual graph is conceptually cleaner and more efficient in route planning than alternative, currently used ways to deal with turn costs. The discussed applications are from the field of pedestrian navigation, which gave rise to this research.

Keywords: shortest path algorithm, turn costs, turn restrictions, route planning, pedestrian navigation, location-based services

1. Introduction

In this paper a model for costs related to the continuation of travel in a node, called turn costs, is investigated. Turn costs require some effort to be treated in representation and analysis of travel networks. When a traveler is reaching the end of an edge, she makes a selection among continuing edges before she proceeds. The selection can be made based on different criteria. For instance, she can select the edge that requires the least angle turn, the edge with the least costs in terms of waiting time, or the edge which continues the actual street. All these examples show turn costs describing binary relationships between (consecutive) edges of a graph. That means, such costs cannot be stored with the edges or nodes in the graph directly.

The relationship can be stored in extra tables if the turn costs cannot be calculated on the fly from geometry. Current navigation data standards use exception tables to represent turn restrictions [14], [25]. Using such extra tables in route finding algorithms would decrease performance. Furthermore, the existence of relations like turn restrictions requires

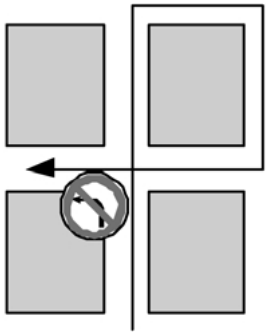


Figure 1. A turn restriction can enforce three right turns instead of one left turn. This route crosses the center node twice.

modifications of the graph or the shortest path algorithms: turn restrictions lead to shortest routes that may include cycles (figure 1), which are not found by shortest path algorithms.

The current way to deal with turn costs is a graph modification. Nodes are split and expanded into elements for each existing binary relation between incident edges [24], [30]. This is not only blowing up the size of the graph significantly, it also introduces a second type of edges with attributes other than the original ones. Because the critical performance factor for shortest path algorithms is the graph size, the need for more elegant solutions puts pressure behind the development of better graph models [5].

Another way of representing turn costs is by a pseudo-dual graph. A pseudo-dual graph replaces edges by pseudo-dual nodes, and turn relations by pseudo-dual edges. It keeps only the topology of the graph, and is significantly smaller than a node-expanded graph. Furthermore, it is a cleaner model, separating nodes (with travel costs) from edges (with turn costs). Shortest path algorithms will run on it without modifications.

This idea was originally proposed by Caldwell [4]. However, it has not found much attention in the literature and was not systematically investigated up to now. In this paper, the pseudo-dual graph is defined in a new way. It is compared with current implementations, and some practical applications are shown with varying severity being handled in other representations. The applications are from the field of pedestrian navigation. The pseudo-dual graph proves to be versatile and efficient for modeling turn costs.

The paper starts with a motivation for modeling turn costs as a central requirement of navigation services (Section 2). Then a formal definition of a pseudo-dual graph, routes in a pseudo-dual graph, and cost functions follows (Section 3). The pseudo-dual graph is then investigated in terms of size, and compared to the node-expanded graph (Section 4). In Section 5 different uses of the turn cost functions are discussed. Some of them are implemented (Section 6). The paper closes with a discussion (Section 7).

2. Route planning with turn restrictions

Route planning is a significant part of navigation that consists of planning, instructing, and acting, i.e., moving in an environment [18], [23], [26]. It depends largely on the *a-priori*

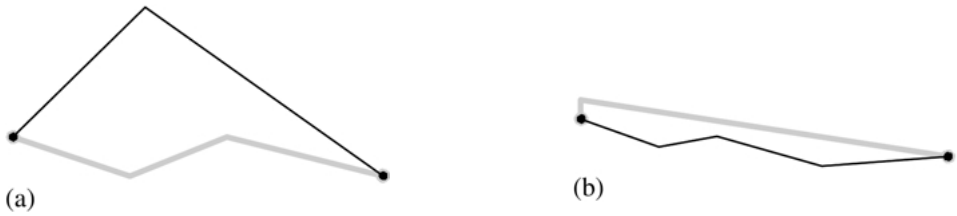


Figure 2. (a) The simplest route (black) may be longer than other routes including more turns (gray). (b) The route of minimal angle (black) may be longer than other routes (gray).

knowledge of the environment, the (mental) map of the planner. For the problems considered here we assume that the knowledge of the environment is complete, i.e., the complete navigable network is known. This is the perspective of an expert, for instance, of a route planning service. The paper does not deal with exploration and construction of mental maps [12], [19], nor with wayfinding strategies of human beings that have incomplete knowledge [20].

Route planning of services results in instructions to users. Only the user demands determine the quality of the service. Hence, human users' perception of urban environments, their conceptualization of the world into objects, and their own moving activities have to be mapped into algorithms and forms of communication [8], [10], [16], [17]. Modeling some of these concepts with the pseudo-dual graph will help to improve navigation services.

Turns are interesting properties of routes, representing their discontinuities. Their significance qualifies them to be used in route descriptions. Assuming that for each turn an instruction has to be given, each turn is a complication of a route description. With this correlation, a minimal link route is the one with the shortest (simplest) description. In some contexts users demand the simplest route (figure 2(a)).

A turn can also be perceived as an enforced deviation from the current direction. Hence, the route of minimal deviations from a global direction may be perceived as optimal (figure 2(b)), because users feel more comfortable if they do not change the direction too much.

Turn costs are also relevant in many other contexts: costs per turn as paid by car drivers for slowing down and accelerating again, costs by legal rules, like restrictions or fees, waiting time for a connection, turn radius or other mobility constraints as relevant, for instance, for long vehicles, or costs proportional to angles. Some of these functions will be illustrated by the applications in Section 5.

3. The pseudo-dual graph

In this section graph concepts used in this paper are recapitulated. The pseudo-dual graph is formally introduced. Concepts of a route in a pseudo-dual graph, and a cost function are

also provided. The latter covers edge costs as well as turn costs, and allows thereby to skip the original graph from consideration after construction of the pseudo-dual graph.

3.1. Graphs

A graph G consists of a set of nodes N and a set of edges E. The edges are defined as pairs of nodes, distinguishing start nodes f: $E \to N$, and end nodes l: $E \to N$. Undirected edges are treated here as a pair of bi-directional edges. Typically edges have some costs attached, w: $E \to \mathbb{R}^+$.

$$G = G(N, E) \quad \text{with } f_G, l_G, w_G.$$

We call edges (e_i, e_j) consecutive if $l(e_i) = f(e_j)$. If the real costs c for traveling two consecutive edges e_i, e_j are greater than $w(e_i) + w(e_j)$, then also the turn in the common node contributes to c. Therefore, we consider another cost function w: $E \to E \to \mathbb{R}^+$ for the turn costs. Note that in a graph G turn costs cannot be attached as an attribute to any of its types of elements.

The total number of edges ending in a node n is called the *indegree* of n, $\delta^+(n) = \sum_{i=1}^{|E|} e_i \mid l(e_i) = n$. The number of edges starting in n is its *outdegree* $\delta^-(n) = \sum_{i=1}^{|E|} e_i \mid f(e_i) = n$. The degree of a node is $\delta = \delta^+ + \delta^-$.

A route r from node $f(e_1)$ to $l(e_k)$ is the sequence of pairwise consecutive edges $r = (e_1, \ldots, e_k)$. Routes may visit a node more than once, which forms a cycle. A route with no cycle is a path. If $f(e_1) = l(e_k)$ the route is called a tour. The length of a route is k; the total costs of a route are $W(r) = w(e_1) + \sum_{i=2}^{k} (w(e_{i-1}, e_i) + w(e_i))$. An optimal path is a route of minimal costs between a given source and target node.

Many real world route problems deal with mobile agents starting on and targeting to intermediate positions along edges, instead of graph nodes. This special case is usually reduced to the general problem by introducing temporary source and target nodes at the start and target position, splitting the start and target edge into two temporary edges.

3.2. Definition of the pseudo-dual graph

Given a graph $G(N, E)$—which will be called the *primal* graph for better distinction—its pseudo-dual graph is defined as follows:

Definition: The graph $D(N_D, E_D)$ with f_D, l_D, w_D is called the *complete pseudo-dual graph* of $G(N, E)$ with f_G, l_G, w_G if:

- For each edge $e_i \in G$ there is a node $v_i \in D$ with $v_i = d(e_i)$ and d being bijective so that $d^{-1}(v_i) = e_i$. Thus, $N_D = d(N_G)$.

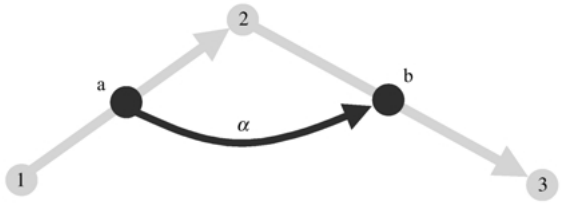


Figure 3. A primal graph G (gray) and its pseudo-dual graph $D(G)$ (black).

- For each pair of consecutive edges $(e_i, e_j) \in G$ there is an edge $\varepsilon \in D$ between the corresponding nodes $v_i = d(e_i), v_j = d(e_j)$, such that $f(\varepsilon) = v_i$ and $l(\varepsilon) = v_j$. $E_D = \bigcup_i \varepsilon_i$.
- There is a cost function w^e: $E_D \rightarrow \mathbb{R}^+$, and a cost function w^v: $N_D \rightarrow \mathbb{R}^+$.

Consider figure 3: an edge α in D represents a path in G from an edge a to a consecutive edge b. The cost $w^e(\alpha)$ could, for instance, be the angle between a and b. Note that in the pseudo-dual graph both cost functions can be attached as attributes to graph elements.

Consider a primal graph as shown in gray in figure 4(a). In this figure, each undirected primal edge corresponds to a directed forth and back edge, which is not shown. The complete pseudo-dual graph contains edges that permit turns on primal edges without passing the end nodes. Such behavior may be legal in some contexts (e.g., pedestrian movement), but it has to be avoided in other contexts. For instance, car drivers should never be advised to drive through a street, then turn and drive back. For such cases, the considered pseudo-dual edges can be excluded, receiving a restricted pseudo-dual graph (figure 4(b)):

Definition: A *restricted pseudo-dual graph* R is a complete pseudo-dual graph D without edges that connect primal alternating consecutive edges: $E_R = \{\varepsilon \in D \mid l(e_i) = f(e_j) \wedge f(e_i) \neq l(e_j)$ for $e_i = d^{-1}(f(\varepsilon)); e_j = d^{-1}(l(\varepsilon))\}$.

The pseudo-dual graph is a graph, and it preserves connectivity. Generally, the pseudo-dual graph of a pseudo-dual graph is not the primal graph (exceptions exist for some

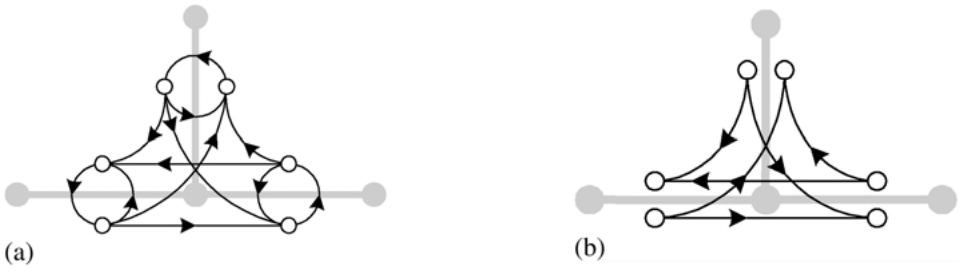


Figure 4. The complete pseudo-dual graph (a, black) of a primal graph that is shown simplified with undirected edges, and the restricted counterpart (b, black).

graphs with a mean degree of 2). However, the property $D(D(G)) = G$ would be one of the requirements of duality for functions in category theory [3]. For that reason we speak of a *pseudo-dual* graph here (Caldwell [4] calls it a *pseudo-network*, and Anez et al. [1] use wrongly the term *dual* graph for the same concept). Also the term *linear dual* graph, as used by the author in earlier papers [27], [28], is misleading.

3.3. Routes in the pseudo-dual graph

Given a primal route $r = (e_1, \ldots, e_k)$ from a source node s to a target t, the corresponding elements in the pseudo-dual graph are $(v_1 = d(e_1), \ldots, v_k = d(e_k))$ and the connecting pseudo-dual edges $(\varepsilon_1, \ldots, \varepsilon_{k-1})$ such that $v_1 = f(\varepsilon_1), v_2 = l(\varepsilon_1) = f(\varepsilon_2), \ldots, v_k = l(\varepsilon_{k-1})$. Semantically, the pseudo-dual source v_1 and target v_k refer to primal edges; the links to the primal source and target are not explicit and need to be determined by functions $s = f(d^{-1}(v_1))$ and $t = l(d^{-1}(v_k))$.

The first edge of any route from s to t is necessarily one of the outgoing edges E_s of s, $E_s = \{e_i \in E \mid f(e_i) = s\}$. Similarly, the last element is one of the incoming edges of t, $E_t = \{e_i \in E \mid l(e_i) = t\}$. That means the route finding problem in the pseudo-dual graph deals with many sources $\Upsilon_s = d(E_s)$, the *source gateways*, and many targets $\Upsilon_t = d(E_t)$, the *target gateways*. The optimal route will lead from any source gateway to any target gateway. Consider figure 5: A route from node 1 to 5 can start with edge a only, but it can end with one of the edges f or c. Thus, the source gateway is $\{d(a)\}$, and the target gateways are $\{d(f), d(c)\}$.

The problem can be reduced to a single-source/single-target problem by adding a virtual source node and a virtual target node to D. The virtual source node v_s is connected to all gateways in Υ_s by virtual edges Φ_s, and the target gateways Υ_t are connected to the virtual target node v_t by virtual edges Φ_t. With the virtual extensions, a primal route $r = (e_1, \ldots, e_k)$ has a pseudo-dual route $(v_s, v_1 = d(e_1), \ldots, v_k = d(e_k), v_t)$, or, in terms

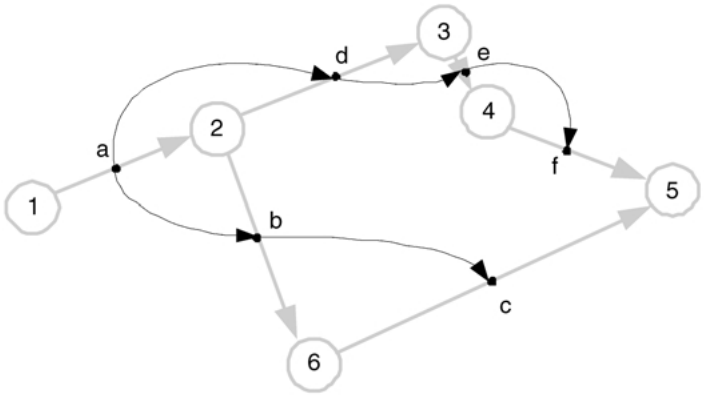


Figure 5. A pseudo-dual route problem may have one or more candidates for source and target nodes.

of pseudo-dual edges, $\rho = (\varepsilon_s, \varepsilon_1, \ldots, \varepsilon_{k-1}, \varepsilon_t)$. Hence, the length of a pseudo-dual route is always greater than the length of the primal route: $|\rho(\varepsilon)| = |r(e)| + 1$.

3.4. Cost functions for the pseudo-dual graph

Analog to the total costs of the primal route the total costs of the pseudo-dual route are:

$$W(\rho) = w^{\upsilon}(\upsilon_s) + w^{\varepsilon}(\varepsilon_s) + w^{\upsilon}(\upsilon_1) + \sum_{i=1}^{k-1}(w^{\varepsilon}(\varepsilon_i) + w^{\upsilon}(l(\varepsilon_i))) + w^{\varepsilon}(\varepsilon_t) + w^{\upsilon}(\upsilon_t).$$

The virtually introduced nodes and edges do not contribute to costs, $w(\upsilon_s) = w(\upsilon_t) = 0$, and $w(\varepsilon|\varepsilon \in \{\Phi_s, \Phi_t\}) = 0$. That way the total costs of a route are not changed by the extension:

$$\begin{aligned} W(\rho) &= w^{\upsilon}(\upsilon_1) + \sum_{i=1}^{k-1}(w^{\varepsilon}(\varepsilon_i) + w^{\upsilon}(l(\varepsilon_i))) \\ &= W(d(\rho)). \end{aligned}$$

Looking for an optimal route requires consideration of the costs traveling primal edges, w^{υ}, as well as the costs turning between consecutive primal edges, w^{ε}. Therefore, another cost function is attached to pseudo-dual edges combining the two costs:

$$w(\varepsilon_i) = w^{\varepsilon}(\varepsilon_i) + w^{\upsilon}(f(\varepsilon_i)).$$

This cost function shortens the formula for the total costs of a pseudo-dual route:

$$W(\rho) = w^{\upsilon}(\upsilon_s) + w(\varepsilon_s) + \sum_{i=1}^{k} w(\varepsilon_i) + w(\varepsilon_t).$$

The short form still covers all costs of a route. With $w^{\upsilon}(\upsilon_s) = 0, w(\varepsilon_s) = w^{\upsilon}(\upsilon_1)$, $w(\varepsilon_t) = 0$, and setting $s = 0$ it can even be shortened to its final form:

$$W(\rho) = \sum_{i=0}^{k} w(\varepsilon_i).$$

In summary, the construction of the pseudo-dual graph needs the primal graph. Afterwards, neither the definition of a route nor any cost function needs to go back to the primal graph.

4. Storage requirements

The storage requirements of the pseudo-dual graph are compared with the widely used node expansion method. It is shown that the pseudo-dual graph is more effective in typical route planning applications in the geographic domain.

4.1. Size of the pseudo-dual graph

The size of the complete pseudo-dual graph D is given by the following equations:

- The number of nodes in D equals the number of edges in G : $|N_D| = |E_G|$.
- The number of edges in D equals the number of paths of length 2 in G. For each $n_i \in N_G$ there are $p(n_i) = \delta^+(n_i) \times \delta^-(n_i)$ such paths. Hence, for a primal graph G the pseudo-dual graph D contains $\sum_{n_i \in N_G} p(n_i)$ edges.

These exact formulas can be estimated, either roughly or rigidly. A rough estimation uses reasonable assumptions for the mean degrees in traffic networks. In the geographical domain, traffic networks are typically sparse. For instance, with mean degrees of $\delta^+ = \delta^- = 3$ the number of linear dual edges is about $9|N_G|$ or $(O(n))$. A rigid upper estimation is based on the Cauchy-Schwarz inequality, writing the sum of the products of the in- and outdegrees as a vector product:

$$\begin{pmatrix} \delta_1^+ \\ \cdots \\ \delta_n^+ \end{pmatrix}^T \begin{pmatrix} \delta_1^- \\ \cdots \\ \delta_n^- \end{pmatrix} \leq \left| \begin{pmatrix} \delta_1^+ \\ \cdots \\ \delta_n^+ \end{pmatrix} \right|_x \left| \begin{pmatrix} \delta_1^- \\ \cdots \\ \delta_n^- \end{pmatrix} \right|_y$$

with $\frac{1}{x} + \frac{1}{y} = 1$. Choosing for x the L_1-norm and for y the L_∞-norm yields $(\sum_i \delta_i^+)(\max(\delta_i^-))$. Assuming that $\max(\delta_i^+) = \max(\delta_i^-)$ the choice is symmetric. Additionally, it holds that $(\sum_i \delta_i^+) = (\sum_i \delta_i^-) = |E_G|$, such that the above inequation yields $|E_D| \leq |E_G| \min(\max \delta^+, \max \delta^-)$, which again is $\leq |E_G| \frac{\delta_{\max}}{2}$. That means, both $|N_D|$ and $|E_D|$ are linear with $|E_G|$.

4.2. Size of node expansion

The common way to handle turn costs, e.g., turn restrictions, in current network analysis tools is *node expansion* [1], [30]. By node expansion, each network intersection blows up to individual representations of all possible connections. The number of nodes increases as well as the number of edges (figure 6):

- Any node n expands into $\delta(n)$ nodes.

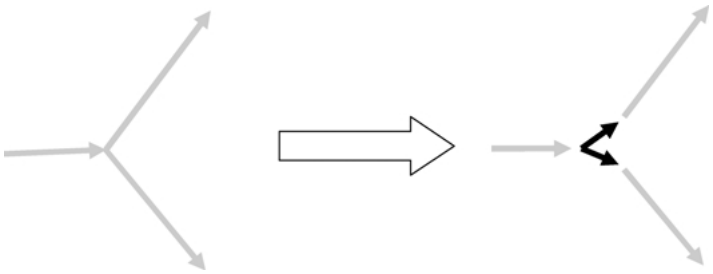


Figure 6. Node expansion.

- For each path of length 2 through n, an additional edge is inserted. The original edges are preserved.

The inserted edges can be interpreted as the travel from entering a complex intersection till leaving [24]. Turn costs can be attached to the inserted edges, which solves the problem of representing the relation attribute between two consecutive edges. However, in a node expanded graph there exist two different types of edges, one with travel costs, and one with turn costs.

The size of a node-expanded graph G_e can be estimated to:

- $|N_{G_e}| = \delta_{\max}|N_G|$.
- $|E_{G_e}| \leq |E_G| + |E_G|\frac{\delta_{\max}}{2} = (1 + \frac{\delta_{\max}}{2})|E_G|$.

4.3. *Comparison of storage requirements*

The comparison of storage requirements shows that both approaches are linear in the number of elements compared to the primal graph. However, for the same primal graph G, in both categories, nodes and edges, the size of the node-expanded graph G_e exceeds the size of the pseudo-dual graph D (table 1). With $\delta_{\max}|N_G| \geq 2|E_G|$ the number of nodes in G_e is at least twice the number of nodes in D. The difference in the number of edges is smaller.

Provided that both approaches allow to store and analyze turn costs, which information is then lost in the pseudo-dual graph, such that it can be significantly smaller? The answer to this question is: the pseudo-dual graph represents the complete topological information,

Table 1. Estimation of upper limits for node-expanded and pseudo-dual graph.

	G_e	D
$\|N\|$	$\delta_{\max}\|N_G\|$	$\|E_G\|$
$\|E\|$	$(1 + \frac{\delta_{\max}}{2})\|E_G\|$	$\frac{\delta_{\max}}{2}\|E_G\|$

but it is not rich enough to be embedded in space. Just to follow the direction of primal edges many intermediate nodes and edge splittings were necessary. The node-expanded primal graph has preserved this metric property. Hence, the pseudo-dual graph is rich enough for route planning, but it is insufficient (and needs reference to the primal graph) for route drawing.

Besides the storage requirements, the observation holds for computation efficiency also. An optimal route algorithm is quadratic [11] or at least of $O(n \log n)$ with the number of nodes [7]. Hence, the computation of an optimal route is more than linearly faster with the pseudo-dual graph.

5. Applying costs to turns

In Section 3 it was shown that primal optimal route problems can be converted into equivalent pseudo-dual optimal route problems. Adding a turn cost function even increases the modeling flexibility. We discuss here three different cost functions by examples: the classical one of car navigation in the presence of turn restrictions [4], navigation of pedestrians and their abstracted perceptions of turns, and navigation with the minimal amount of turn angles. Then we apply the pseudo-dual graph for its topological properties in a pedestrian route planning task.

5.1. Routes with turn restrictions

Consider a street network with some turn restrictions for car drivers. A typical situation was shown in figure 1: to reach the target, the driver crosses the intersection with the left turn restriction, turns right three times and crosses the same intersection, now in the orthogonal direction. It was mentioned in Section 1 that Dijkstra's algorithm cannot find such routes in the primal graph. In a restricted pseudo-dual graph turn restrictions can be modeled by assigning infinite costs to the corresponding pseudo-dual edges. It can easily be verified that, in the situation shown, no node in the pseudo-dual graph needs to be visited more than once. The resulting route is highlighted in figure 7.

5.2. Turn concepts of pedestrians

The second application considers the navigation of pedestrians in urban space and their abstracted perception of turns that can be modeled directly in terms of discrete turning costs.

Nodes appear in a street network for many reasons, and not all nodes are perceived as turns. Some intermediate nodes separate two network segments because of a change of a street attribute, although the direction of the street does not change. Other nodes describe a change of direction but offer no alternative than following a route. Higher level concepts

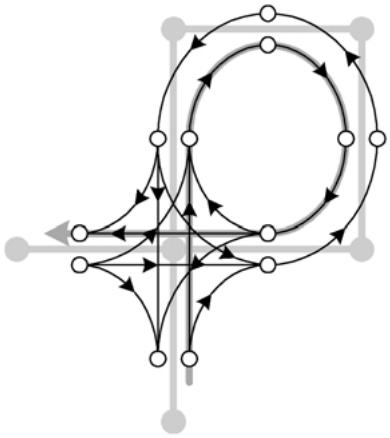


Figure 7. The pseudo-dual graph for the situation in figure 1.

of turns come into play. Pedestrain guidance services could use such higher level concepts to control instructions given to the pedestrians.

In the following examples the situation shown in figure 5 is used for explanation. In each case node 1 is considered as source node, and node 5 is the target node. Starting from node 1, there are two different routes to node 5: $r_1 = (a, b, c)$ and $r_2 = (a, d, e, f)$, or $\rho_1 = (\beta, \delta)$ and $\rho_2 = (\alpha, \varepsilon, \zeta)$ respectively.

- Route of minimal number of turns.

  The route of the minimal number of turns can be determined by weighting all pseudo-dual edges ε with 1. That is equivalent to counting edges in G and does not justify the effort for constructing the pseudo-dual graph. However, the concept can later be modified to be more useful. For the present we set the costs for the pseudo-dual graph in figure 5 to:

  $$w(\alpha) = 1,\ w(\beta) = 1,\ w(\delta) = 1,\ w(\varepsilon) = 1,\ w(\zeta) = 1$$

  In this case, the optimal route is $\rho_1 = (\beta, \delta)$ with total costs of $W = 2$.

- Generalizations of turns.

  Human perception generalizes. An angle between two consecutive primal edges may be perceived as a turn if it is above a certain threshold. Small angles are ignored. This could be modeled by the following edge costs:

  $$w(\alpha) = 0,\ w(\beta) = 1,\ w(\delta) = 1,\ w(\varepsilon) = 1,\ w(\zeta) = 1$$

In this case, both routes in figure 5 are of the same total costs $W = 2$; the choice of the algorithm is not deterministic.

- Semantic concepts of turns.

  At the semantic level, pedestrians could perceive a turn when leaving one street for turning into another. Consider an instruction like: ''Follow the Broadway until ...''. With this instruction movement continues until the Broadway has to be left, even if, while moving along the Broadway, the current orientation would change. Applying such a concept to figure 5, we add the knowledge that a and d are segments of *A-Street*, and e and f are segments of *B-Street*. Then the costs are:

  $$w(\alpha) = 0,\ w(\beta) = 1,\ w(\delta) = 1,\ w(\varepsilon) = 1,\ w(\zeta) = 0,$$

  The cheaper route is ρ_2: $W(\rho_2) = 2$, and $W(\rho_1) = 3$.

5.3. *Route of minimal angle*

In the third application we search for routes of the minimal total turning angle. Continuing a route straight causes no costs, but making a peak turn causes high costs:

$$w(\alpha) = 0,\ w(\beta) = \frac{\pi}{2},\ w(\delta) = \frac{\pi}{2},\ w(\varepsilon) = \frac{\pi}{2},\ w(\zeta) = \frac{\pi}{4},$$

The cheaper route is route ρ_2 with costs of $W(\rho_2) = \frac{3}{4}\pi$, in comparison to $W(\rho_1) = \pi$. In a Manhattan-like grid, the optimal route will be the route with maximal legs. This behavior of the algorithm differs from the least angle heuristics [20] which tries to keep the globally known direction and would follow a zigzag route in the same environment.

5.4. *Specifying route properties*

Route planning is not always a (geometric) shortest path problem; for instance, planning hiking trips involves quite different criteria on the properties of routes. The above discussed properties of pseudo-dual routes can also be used for the specification of desired topological route properties.

A circular tour is a popular hiking trip type; imagine hikers who want to return to their car at the end of the trip. In the primal graph, a shortest path algorithm will stop after initialization. In the pseudo-dual graph start and target gateways are disjunct, and thus, the length of a circular tour is at least 2 (figure 8(a)). Varying the initial solution by k-shortest path algorithms [2], [21] would provide longer circular tours also.

Hiking routes may include cycles in between, a property of pseudo-dual routes discussed in Section 5.1 already. Hiking routes may include segments to be traveled in



Figure 8. Round trips (a) and even segments being traveled in both directions (b) do not require visiting a pseudo-dual node more than once.

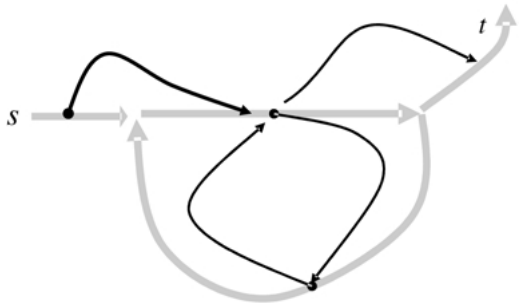


Figure 9. If a segment shall be traveled twice, a pseudo-dual node would be visited twice.

both directions: no pseudo-dual node has to be visited more than once (figure 8(b)). In contrast, nobody wants to hike the same segment twice in the same direction. That is excluded in the pseudo-dual graph, meaning that a pseudo-dual node would be visited twice (figure 9).

Thus, the desired route properties turn out to be still paths in the pseudo-dual graph, where unwanted properties are prevented. That means that cycle-free k-shortest path algorithms [6], [29] applied on a pseudo-dual graph yield routes of the desired properties automatically.

On the primal graph, cycle-free k-shortest path algorithms produce none of the above shown desired route shapes. Other k-shortest path algorithms allowing cycles [13] run on the primal graph into combinatorial explosion, being not limited by the above discussed constraints.

6. Implementations and tests

The pseudo-dual graph is formally specified for testing its correctness and its properties. Additionally the pseudo-dual graph is applied in a route planning tool for hiking trips. In this section we report on both.

6.1. *Functional specification*

The specification of the pseudo-dual graph is written in a functional programming language, which provides an executable prototype. The prototype was tested for the correctness of the described model. Further it allowed to check the properties, like amount of storage, or the behavior of specific cost functions, with real world test data sets. The sample data set contained an essential part of the traffic network in Vienna, about 2,500 streets, including turn and one-way restrictions. The behavior of the prototype was as expected.

The language used for the specification is Haskell [22]. The full source code can be downloaded at ftp://ftp.geoinfo.tuwien.ac.at/winter/winter02modeling-code.zip. In the code, Dijkstra's classical shortest path algorithm is used based on priority queues, taking ideas from the Functional Graph Library [15]. However, the purpose was to produce clear and easily readable code, so the code is not optimized for performance.

6.2. *Planning hiking trips*

A route planner for hiking tips was implemented in Java by Cziferszky [9]. The program generates the pseudo-dual graph from a primal graph, and calculates a set of route candidates with the desired topological properties applying a k-shortest path algorithm. The result is an ordered list of k routes from source to target with monotonic increasing costs. The routes are then evaluated for other properties, especially the parameters of the actual request, like length, difficulty, maximal slope, and so on. Only the pseudo-dual graph could provide the topological route properties.

7. Discussion

I presented an approach to model turning costs in networks, based on a pseudo-dual graph. The pseudo-dual graph represents all pairs of consecutive edges and allows individual weighting. Its data structure is simple (basically the same as for the primal graph), its construction is simple (replacing edges by nodes and adding all paths of length 2 by an edge), and shortest path algorithms can be applied unchanged. It has been shown that the pseudo-dual graph is a suitable representation of directed networks for routing purposes.

Although the idea for the pseudo-dual graph goes back to Caldwell [4], it has not found much attention in the literature. Recently, Anez et al. [1] used the idea for modeling change costs in a multi-modal transport network. As they state, no (other) practical applications were developed from Caldwell's proposition. In this paper cost functions are developed that allow to skip the primal graph completely from consideration during route finding. For the first time a systematic investigation of memory requirements is presented. Further, with the exception of turn restrictions, all applications of the pseudo-dual graph are original.

Some examples illustrated the relevance of the approach for practical applications. Most of them argue from a cognitive point of view. Adapting location-based services to people's needs requires options like searching for the most simple route, including concepts what ''simple'' means in a specific domain. For pedestrian navigation in an unfamiliar urban environment, simplicity can be measured, for example, in terms of description length or of minimal angles along the route.

The work is part of the effort to develop better location-based services. Further investigations are needed to test the validity of cognitive assumptions made, to characterize the behavior of the algorithms in extreme cases, and to communicate the route planning strategy to users. Some of the open questions are:

- Do the presented cost functions satisfy real needs? Or are there others that would? Clearly the shown set of applications is not complete. Also not investigated so far is the possibility to combine different cost functions to useful optimization criteria.
- How can a mobile agent be located in the pseudo-dual graph? Although the pseudo-dual graph is topologically complete to deal with route planning, it is not embedded in space up to now. (That is the reason why it is more efficient than node expansion.)
- Can the performance be increased? The implemented prototypes have a poor performance if the typically large networks and multi-user access are considered. A commercial service needs strategies to separate offline- and online-tasks, to maintain the pseudo-dual graph after changes in the primal graph, and to allow personalization in cost functions. Also other strategies, like A^* or hierarchical approaches, could be implemented.

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